

Lab exercise 11

DATABASE WITH AUTO_INCREMENT SEQUENCES

AIM

To create and demonstrate an **AUTO_INCREMENT sequence** for the `VegID` column in the `Vegetable` table, manually populate the `VegID`, and remove the `AUTO_INCREMENT` property using MySQL commands.

ALGORITHM / PROCEDURE

1. Start the MySQL application and select the required database.
2. Create or use the `Vegetable` table containing `VegID`, `VegName`, `Category`, `Price`, and `Quantity`.
3. Modify the `VegID` column and set it as `AUTO_INCREMENT` and `PRIMARY KEY`.
4. Insert vegetable records without specifying `VegID`.
5. Display the table and verify that `VegID` values are automatically generated.
6. Insert a vegetable record by manually specifying its `VegID`.
7. Remove the `AUTO_INCREMENT` property from the `VegID` column.
8. Insert another vegetable record by manually entering its `VegID`.
9. Display the final table and verify the changes.
10. Stop.

Question 1 : Manually Populate VegID

```
INSERT INTO Vegetable
(VegID, VegName, Category, Price, Quantity)
VALUES
(10, 'Spinach', 'Leafy', 28.00, 60);
```

Display:

```
SELECT * FROM Vegetable;
```

```
mysql> SELECT * FROM Vegetable;
```

VegID	VegName	Category	Price	Quantity
1	Tomato	Vegetable	30.50	100
2	Potato	Vegetable	25.00	200
3	Carrot	Root	40.00	80
4	Onion	Vegetable	35.00	150
5	Brinjal	Vegetable	45.50	90
6	Cabbage	Leafy	50.00	70
10	Spinach	Leafy	28.00	60

7 rows in set (0.00 sec)

Question 2: Drop AUTO_INCREMENT

```
ALTER TABLE Vegetable  
MODIFY VegID INT PRIMARY KEY;
```

Now VegID will no longer be generated automatically.

Insert manually:

```
INSERT INTO Vegetable  
(VegID, VegName, Category, Price, Quantity)  
VALUES  
(11, 'Beetroot', 'Root', 42.00, 50);
```

Finally:

```
SELECT * FROM Vegetable;
```

```
mysql> ALTER TABLE Vegetable  
-> MODIFY VegID INT PRIMARY KEY;  
ERROR 1068 (42000): Multiple primary key defined  
mysql> INSERT INTO Vegetable  
-> (VegID, VegName, Category, Price, Quantity)  
-> VALUES  
-> (11, 'Beetroot', 'Root', 42.00, 50);  
Query OK, 1 row affected (0.01 sec)  
  
mysql> SELECT * FROM Vegetable;
```

VegID	VegName	Category	Price	Quantity
1	Tomato	Vegetable	30.50	100
2	Potato	Vegetable	25.00	200
3	Carrot	Root	40.00	80
4	Onion	Vegetable	35.00	150
5	Brinjal	Vegetable	45.50	90
6	Cabbage	Leafy	50.00	70
10	Spinach	Leafy	28.00	60
11	Beetroot	Root	42.00	50

```
8 rows in set (0.00 sec)
```